Ring and 2D Two-Target Bimodality Brief

Focus: what was scanned, where double peaks appear, and which criteria should be used.

Luca-Aligned Shortcut Ring

Configuration used here. The configuration is aligned with the Luca Outlook exchange: ring sites are labelled $1, \ldots, N$, $N = 100$, start $n_0 = 1$, absorbing target $= 51$, shortcut source $= 6 = n_0 + 5$, and shortcut destination $= 52 = 51 + 1$. Thus the shortcut is one-way, $6 \to 52$. Unless stated otherwise, $q = 2/3$, $\rho = 1$, lazy_selfloop, and the displayed reference case uses $\beta = 0.01$.

What Was Scanned

Scan	Parameters varied	Observed double-peak window	Readout
Luca shortcut ring, fixed $N = 100$	β for $K = 2, 4$; $q = 2/3$, $\rho = 1$, $6 \to 52$	$K = 2:\ \beta = 0.0005$–0.05; $K = 4:\ \beta = 0.001$–0.15	$\beta = 0$ gives no shortcut peak. Too-large β collapses the shape toward an early shortcut-dominated peak.
Shortcut ring, fixed $\beta = 0.02$	$N = 10, 12, \ldots, 200$, $K = 2, 4$, antipodal target and shortcut just beyond target	$K = 2:\ N \geq 58$; $K = 4:\ N \geq 72$	Small rings do not separate the fast and slow timescales enough. Larger N creates the late non-shortcut peak.
Shortcut ring, fixed $\beta = 0.01$	$N = 50, 52, \ldots, 300$, $K = 2, 4$	$K = 2:\ N \geq 60$; $K = 4:\ N \geq 74$	The practical minimum size is roughly $N \approx 60$ for $K = 2$ and $N \approx 70$–75 for $K = 4$.
Two-target 1D ring	$L = 31, 51$, $K = 2$, drift $g \in \{-0.6, -0.3, 0, 0.3, 0.6\}$, $\beta \in \{0, 0.03\}$, starts and two target locations	34 classifier-confirmed `double_peak` cases out of 240; 14 without shortcut, 20 with shortcut	Double peaks can occur from target-channel competition alone. The shortcut is helpful but not necessary.
Two-target N-β scan	$N = 30, 35, \ldots, 80$, $\beta = 0, 0.01, \ldots, 0.08$, $K = 2, 4$	$K = 2:\ N = 45$–80, $\beta = 0.01$–0.08; $K = 4:\ N = 50$–80, $\beta = 0.01$–0.08	The two-target mechanism needs enough spatial separation and a channel split; otherwise the result is unimodal, shoulder, or local bump.

Table 1: Existing 1D ring scans already present in the repository.

2D Near/Far Target Bias-Radius Scan

Configuration. This part has also been done, but only as a bounded $\theta = 0$ slice rather than a full angular scan. The grid is 15×9, the start is fixed at $(2, 4)$, the far target is fixed at $(12, 4)$, and the near target is placed east of the start at radius r. The boundary is reflecting attempted-outside-stays. The global east bias moves probability mass from stay to the east step:

$$p_E = q/4 + b, \qquad p_N = p_S = p_W = q/4, \qquad p_{\text{stay}} = 1 - q - b,$$

with $q = 0.7$. The scan uses $r = 2, 3, 4, 5$ and $b = 0, 0.04, 0.08, 0.12, 0.16$, giving 20 cases.

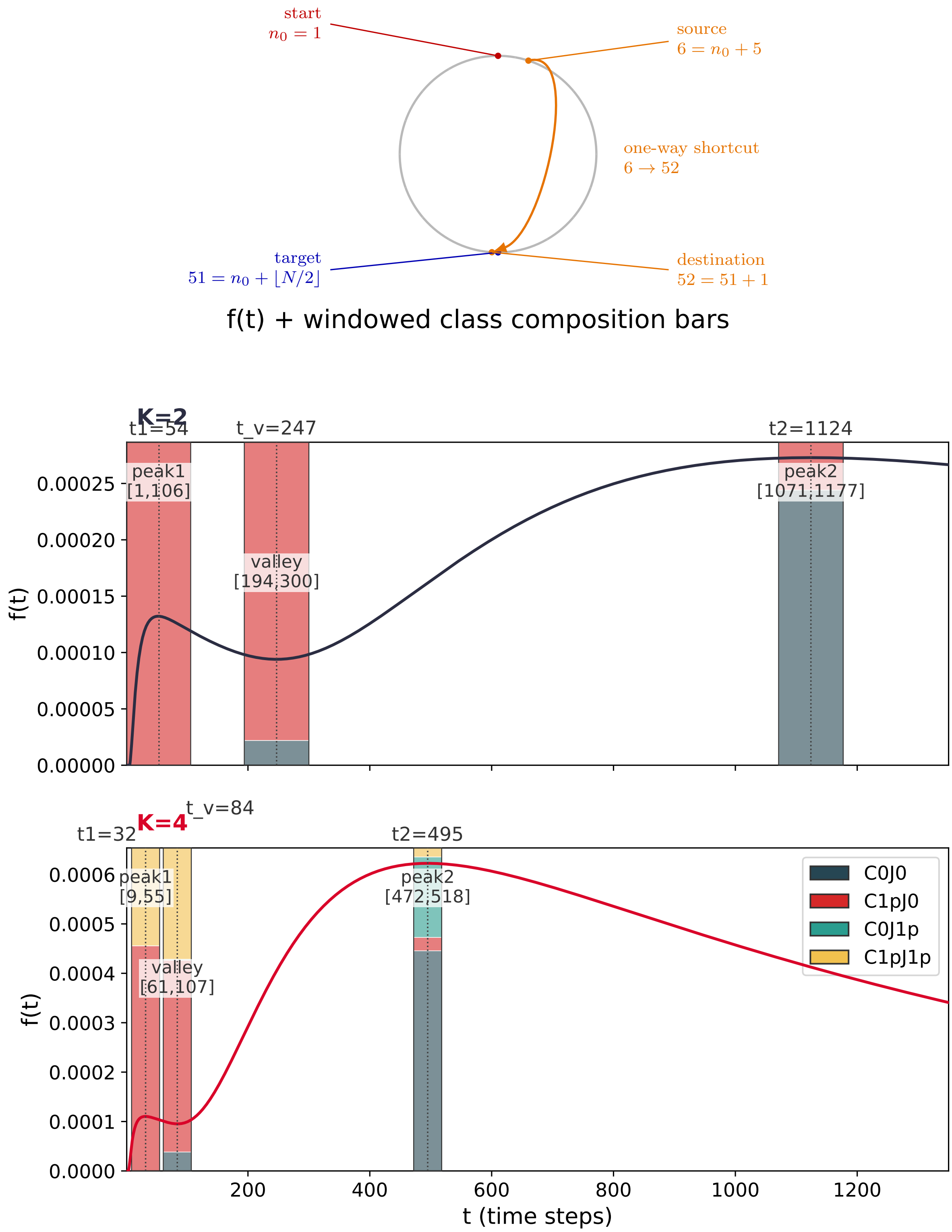

Figure 1: Reference configuration and first-passage curves at $\beta = 0.01$. The early peak is shortcut-driven. The late peak is the slow non-shortcut budget; for $K = 4$, jump-over/overshoot trajectories add a delayed component.

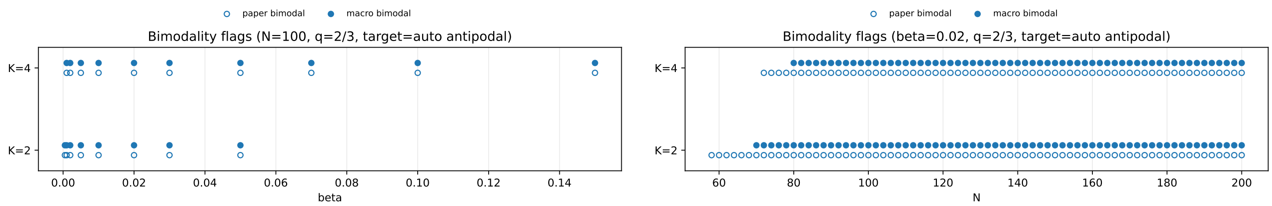

Figure 2: Left: β-scan at $N = 100$. Right: N-scan at $\beta = 0.02$. These figures show where the shortcut-ring classifier finds two separated peaks.

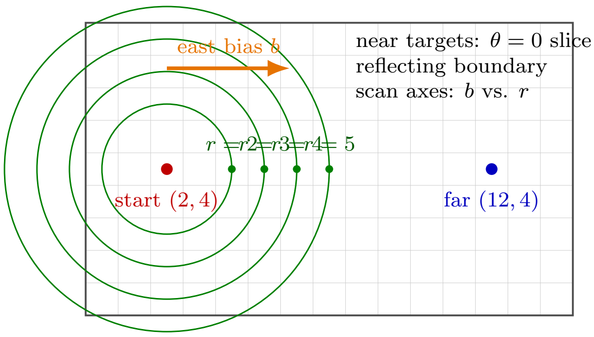

Figure 3: 2D scan geometry. The near target moves eastward from the start, while the far target stays fixed. The heatmap below is therefore a fixed-angle bias-radius slice, not a full angular phase diagram.

Scan	Parameters varied	Readout
2D near/far target, $\theta = 0$	$r = 2, 3, 4, 5$, $b = 0, 0.04, 0.08, 0.12, 0.16$; start $(2, 4)$, far $(12, 4)$	Double peaks appear when near-target hits and delayed far-target hits both retain visible mass. Intermediate/strong east bias often helps, but distance alone is not sufficient.

Table 2: Existing 2D bias-radius scan. Positive claims require the stored classifier label to be exactly `double_peak`.

Classifier-confirmed 2D double-peak cases. The scan found 7 `double_peak` cases out of 20:

$$(r, b) = (2, 0.16),\ (3, 0),\ (3, 0.16),\ (4, 0.08),\ (4, 0.12),\ (5, 0.12),\ (5, 0.16).$$

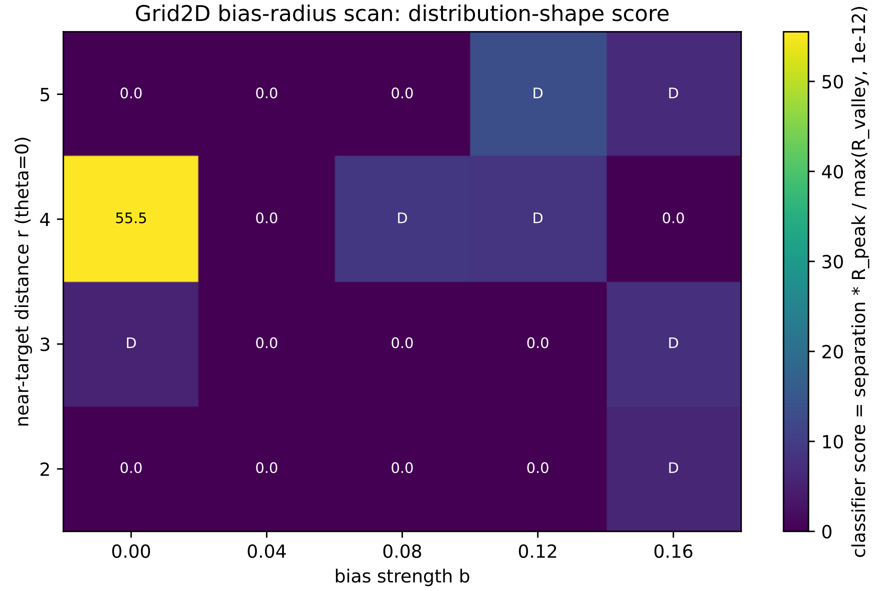

Figure 4: 2D heatmap for the fixed $\theta = 0$ slice. Color is the classifier score, not by itself a double-peak claim. The actual positive cases are the seven rows labelled `double_peak` in the metrics table.

Why the display is binned. The exact computation gives a raw discrete-time PMF on a small grid, so the raw curves contain parity and path-length oscillations, especially near the close target. To avoid a jagged presentation figure, Figure 5 bins probability mass into 4-step windows:

$$F_k = \sum_{t=4k}^{4k+3} f(t).$$

This preserves probability mass and is only a display transform. The scientific claims still use the raw classifier, raw peak times, and near/far channel decomposition.

2D Local-Bias Basin Test

Question. The fixed global-bias two-target scan can produce classifier-confirmed `double_peak` cases, but the visual shape is often weak because the near target is hit by a small number of short paths. I therefore tested a plain 2D rectangle with no corridor geometry, adding only a local bias basin around the near target. The aim is to create a broader near-target residence budget while keeping a delayed far-target budget alive.

Configuration. The grid is a reflecting 23×13 rectangle. The start is $(3, 6)$, the far target is $(19, 6)$, and the far target is aligned with a global east bias. The near target is moved off the east-bias line, e.g. $(5, 9)$ or $(5, 3)$. Around the near target, a local basin of radius R_b is added. Inside this basin there are two local effects:

1. a weak inward bias pointing toward the near target;
2. a local holding/slowing term that increases residence in the basin.

Absorbing targets still take precedence: once either target is hit, the process stops.

Small scan. The first local-bias scan used:

$$R_b \in \{2.5, 3.5, 4.5\}, \quad g \in \{0.08, 0.12, 0.16, 0.20\}, \quad \ell \in \{0, 0.04, 0.08, 0.12\}, \quad h \in \{0, 0.25, 0.45\}.$$

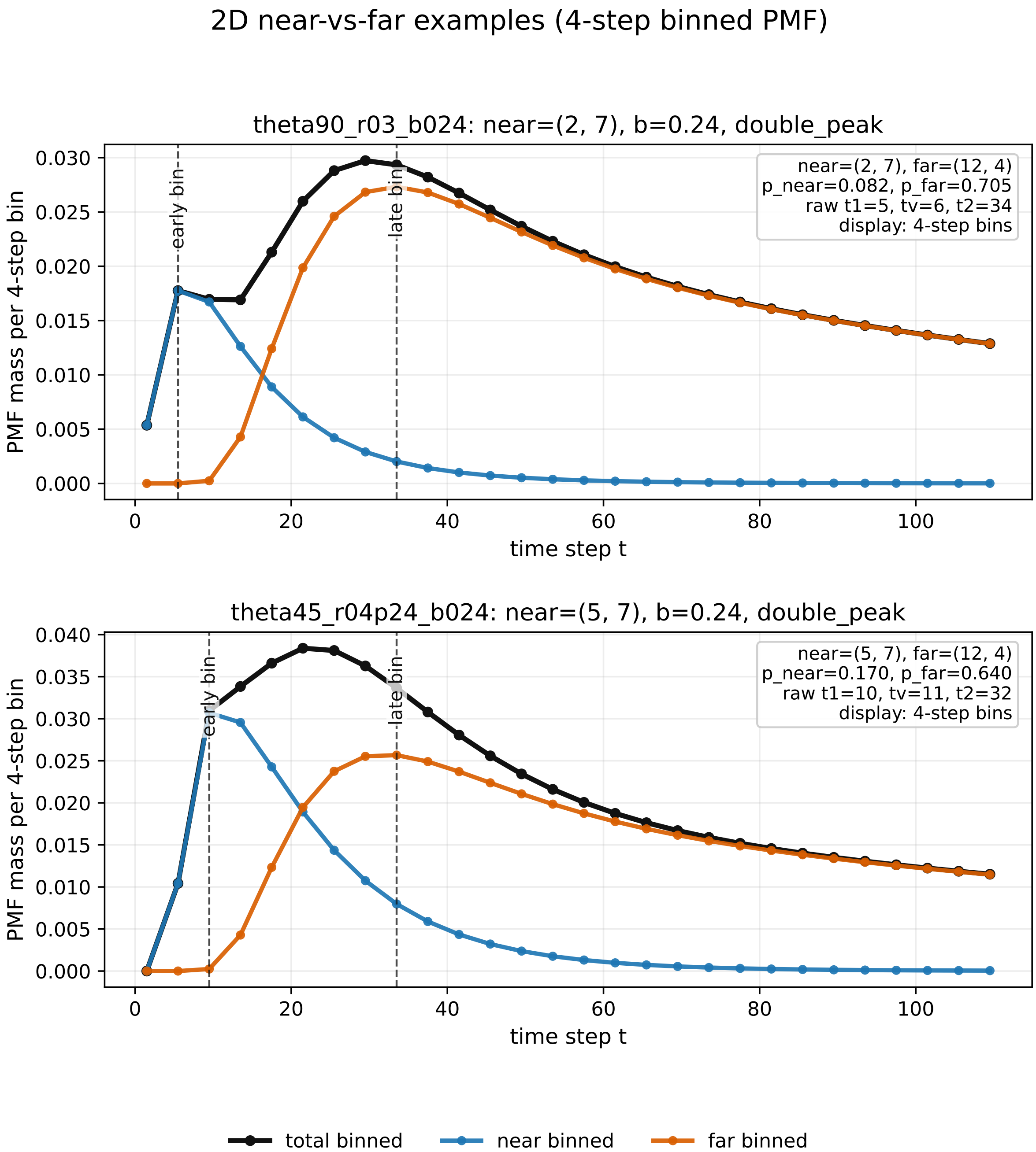

Figure 5: Two clearer 2D double-peak examples from a targeted θ-expanded search, displayed as 4-step binned probability mass for readability. The classifier labels and peak times still come from the raw exact discrete-time PMF. In both cases the near target is off the east-bias line, so the early peak is a near-target channel and the late peak is dominated by delayed arrival at the far target. These examples are illustrative, not a full angular phase diagram.

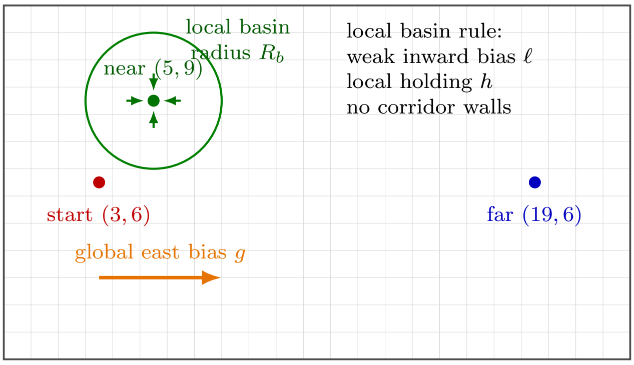

Figure 6: Configuration for the local-bias basin test. The geometry remains a plain reflecting rectangle. The added mechanism is a local near-target basin, not a corridor.

The near targets scanned were $(5, 9), (7, 9), (5, 3), (7, 3)$. This gives 576 cases. The raw exact PMF classifier found 210 `double_peak` cases. Restricting to actual local inward bias $\ell > 0$, it found 126 `double_peak` cases.

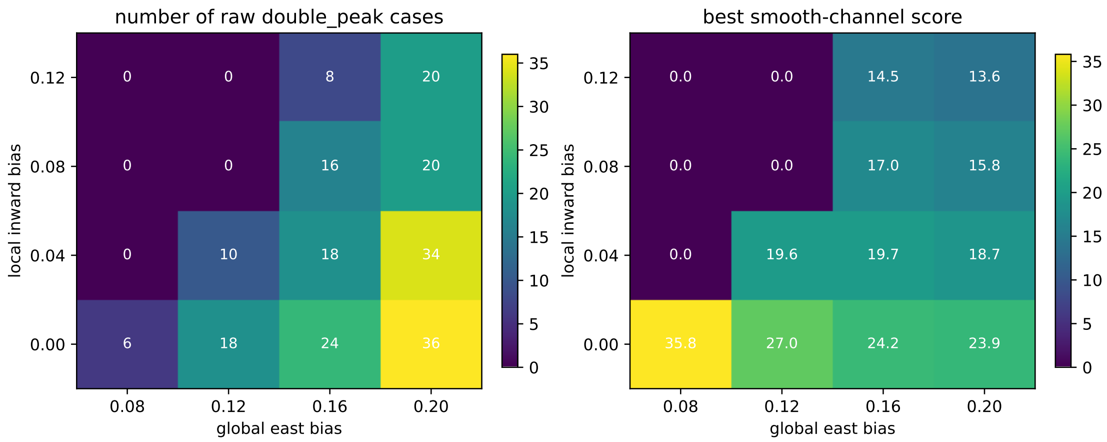

Figure 7: Local-bias basin scan. Left: number of raw classifier-confirmed `double_peak` cases aggregated over near-target position, basin radius, and holding strength. Right: best smooth-channel score used to choose readable examples.

Preliminary conclusion. The useful mechanism is not strong local attraction by itself. Strong local bias tends to make near absorption too fast and sharp. The more promising design is a weak local capture basin plus moderate local residence: global east bias keeps the far-target channel alive, while the local basin broadens the near-target contribution. This produces smoother two-target double-peak-like curves without introducing corridor walls. This is still a pilot scan, not a full 2D local-bias phase diagram.

Useful Double-Peak Criteria

Do not use "two local maxima" alone. That criterion is too permissive: it counts tiny numerical bumps and visually weak shoulders. I would use a two-stage rule.

1. **Candidate screen:** two filtered local maxima above numerical noise, with an inter-peak valley. This corresponds to the existing `paper_bimodal` or `double_peak` classifier rows.
2. **Presentation claim:** report a double peak only when

$$t_2/t_1 \geq 10, \qquad h_2/h_1 \geq 0.05, \qquad h_v/\max(h_1, h_2) \leq 0.6,$$

and the peak pair has a mechanism explanation: shortcut/non-shortcut budget split or target-channel split.

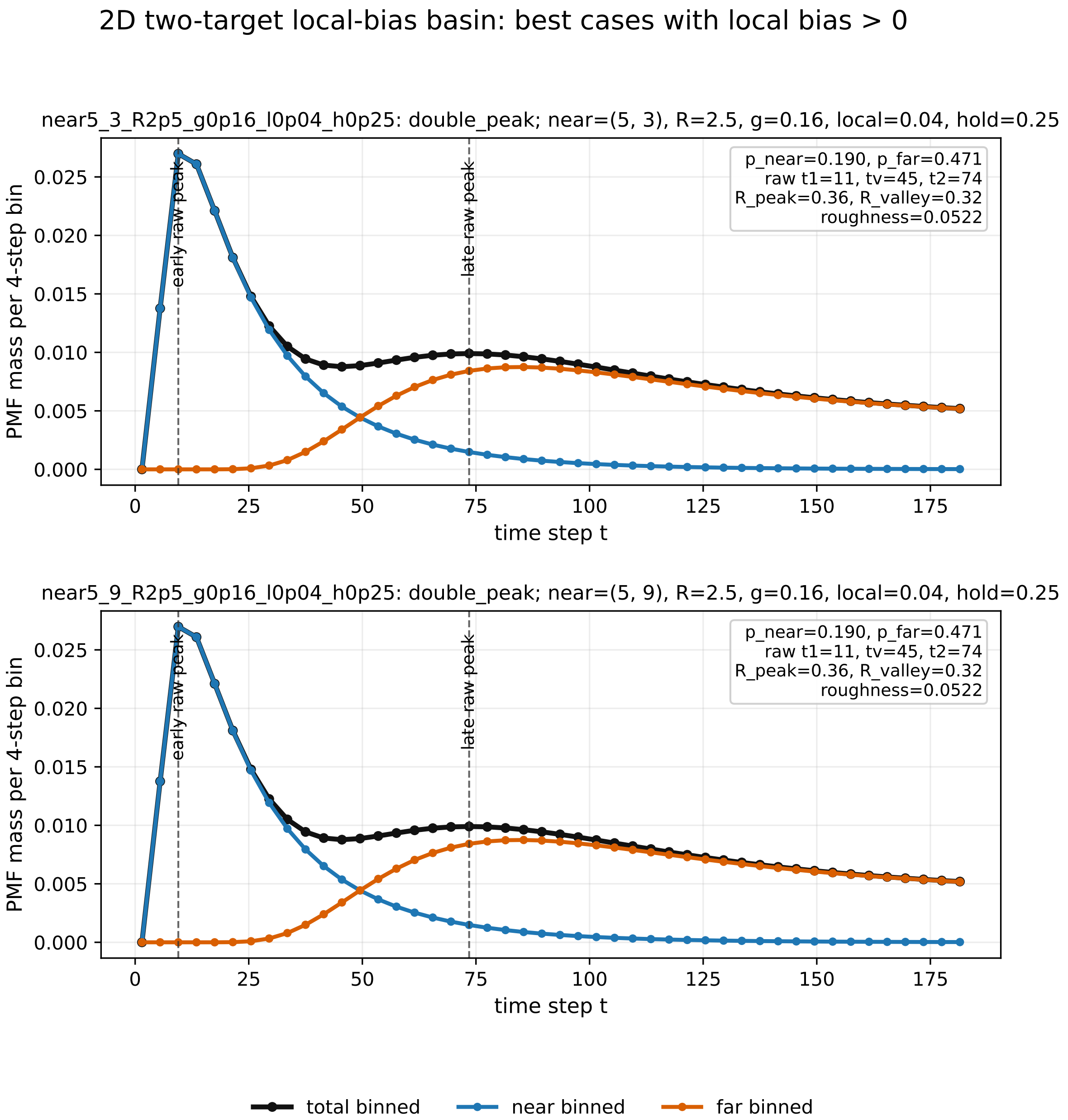

Figure 8: Best local-bias-positive examples, displayed as 4-step binned probability mass. These are still classified using the raw exact PMF. A representative setting is near $(5,3)$ or $(5,9)$, $R_b = 2.5$, global east bias $g = 0.16$, local inward bias $\ell = 0.04$, and local hold $h = 0.25$. The early peak is near-channel dominated, while the later broad bump is far-channel dominated.

3. **Strict visual claim:** if the audience may challenge the shape, use $h_v/\max(h_1, h_2) \leq 0.3$. This is stricter and turns some weak "double peak" candidates into shoulders or local bumps.

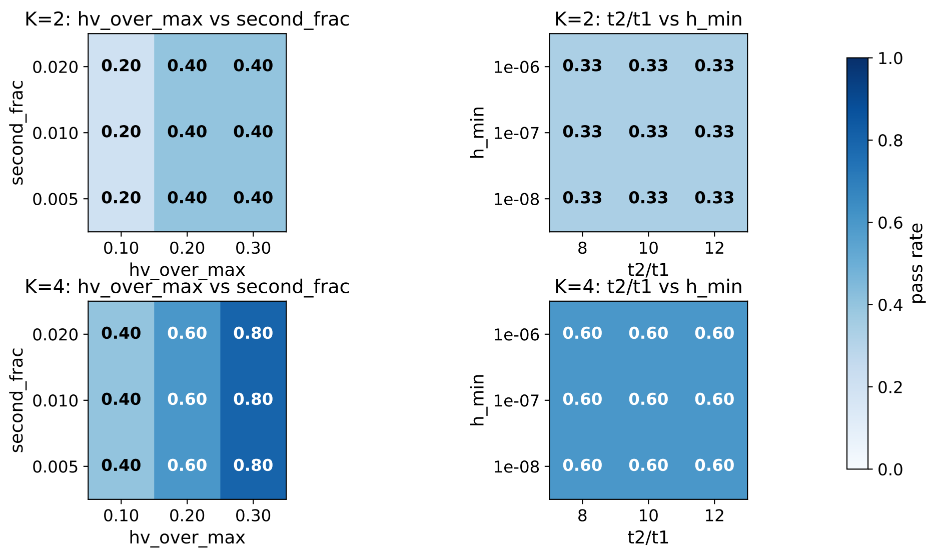

Figure 9: Threshold sensitivity scan. The scan varied the minimum peak height, second-peak fraction, t_2/t_1, and valley-depth threshold. This is why the report should distinguish candidate bimodality from visually clear bimodality.

What Helps or Blocks Double Peaks

Helps. Double peaks are easiest to see when the system creates two arrival budgets with separated timescales. In the Luca shortcut ring this is the fast shortcut budget versus the slow non-shortcut budget. In the two-target ring this is the near/early target channel versus the far/late target channel. In the 2D scan, the same logic becomes a near-target channel versus a far-target channel under eastward bias.

Blocks or weakens. Too small N compresses the two timescales into one peak. Too small β removes the shortcut-driven early budget. Too large β overweights the shortcut route and can erase the late component as a visible second peak. In the two-target scan, weak channel separation gives shoulder or local-bump labels rather than `double_peak`. In the 2D scan, $r = 2$ needs stronger bias to show a labelled double peak, while some larger-radius cases fail if the near and far channels do not both remain visible. The clearer near-vs-far examples found by the targeted θ-expanded search place the near target away from the east-bias line, e.g. near $(2, 7)$ or $(5, 7)$ with $b = 0.24$. This lets the far target keep enough probability mass to form a delayed second peak.

Role of K. For the Luca shortcut ring, $K = 4$ is more robust over β than $K = 2$: the sampled bimodal interval expands from $\beta \leq 0.05$ for $K = 2$ to $\beta \leq 0.15$ for $K = 4$. The mechanism is that $K = 4$ has jump-over or overshoot classes that add a delayed contribution to the slow side of the distribution.

Previously Done but Keep Separate

There is also an older Luca meeting scan for $K = 2$ with fixed shortcut $6 \to 56$ and target 56. That was a useful threshold check, but it is not the same as the later Outlook configuration $6 \to 52$ with target 51. Under that older $6 \to 56$ convention, Luca's $\beta_c = 2q/((1-q)N) = 0.04$ is not an exact boundary for the existence of any two local maxima, but it works as a practical boundary for visually meaningful double peaks: above β_c, the valley becomes too shallow for a clear two-peak claim.

Source Files

- Luca shortcut scans: `ring_lazy_jump_ext/artifacts/data/scan_beta_N100_K2.csv`, `scan_beta_N100_K4.csv`, `scan_N_K2_beta002.csv`, `scan_N_K4_beta002.csv`.
- Larger N-scan at $\beta = 0.01$: `ring_lazy_jump_ext/artifacts/outputs/N_sweep_beta_selected/N_sweep_metrics_beta0.01.csv`.
- Threshold sensitivity: `ring_lazy_jump_ext_rev2/artifacts/outputs/sensitivity/threshold_sweep.csv`.
- Two-target ring: `ring_two_target/artifacts/data/small_scan_metrics.csv`, `scan_bimodality_K2.csv`, and `scan_bimodality_K4.csv`.
- 2D two-target bias-radius scan: `grid2d_two_target_bias_radius/artifacts/data/grid2d_bias_radius_scan_metrics.csv`, `grid2d_double_peak_candidates.csv`, and `grid2d_bias_radius_heatmap_theta0.pdf`.
- 2D targeted examples: `final_multitimescale_fpt_encounter/artifacts/data/grid2d_typical_double_peak_cases.csv` and `grid2d_typical_double_peak_cases.pdf`.
- 2D local-bias basin pilot: `final_multitimescale_fpt_encounter/code/scan_grid2d_local_bias_basin.py`, `final_multitimescale_fpt_encounter/artifacts/data/grid2d_local_bias_basin_scan.csv`, `grid2d_local_bias_basin_heatm` and `grid2d_local_bias_basin_local_positive_best.pdf`.